

Deep Learning Model for Detecting DDoS Attacks in HTTP and HTTPS Traffic

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Abstract—Distributed Denial of Service (DDoS) attacks continue to threaten the stability of online platforms by overwhelming servers with excessive traffic, often disguised as legitimate HTTP and HTTPS requests. Traditional detection techniques struggle to distinguish these attacks from normal user activity, especially as adversaries adopt increasingly sophisticated methods. In this work, we extend this challenge to the Internet of Things (IoT) domain and propose a machine learning–based approach for detecting DDoS attacks using the CIC-IoT2023 dataset. Flow-based network features from the dataset were extracted and used to train and evaluate three deep learning models: a Deep Neural Network (DNN), a Convolutional Neural Network (CNN), and a Transformer-based model. To address the dataset’s structural limitations, we introduce two key methodological contributions. First, we merge DoS and DDoS into a unified Flood class, reducing misclassification caused by overlapping statistical patterns. Second, we apply targeted class balancing to correct the extreme imbalance between attack types, which significantly stabilizes model training. Our results show an improved accuracy of 99%, outperforming common baselines reported in existing literature. The findings demonstrate that carefully integrating dataset restructuring, feature normalization, and modern deep learning architectures can substantially enhance DDoS detection performance. Overall, this study highlights the potential of machine learning to strengthen resilience against evolving DDoS threats in both web and IoT environments..

I. INTRODUCTION

This study aims to minimize the prevalence of DDoS attacks by evaluating different methods and models for their detection. The dataset we used, CIC IoT-2023, consists of network traffic data from a variety of common IoT devices. This is an important consideration, as IoT devices are becoming increasingly widespread and vulnerable to malicious attacks. Detecting and preventing these attacks is critical for ensuring the security and privacy of consumers who rely on IoT devices. Researchers have previously employed various deep learning and machine learning techniques to classify network traffic using multiple datasets. Based on the literature review, deep learning models generally achieve the highest accuracy, whereas machine learning models exhibit moderate performance with lower precision. A performance gap remains between machine learning and deep learning approaches. Although deep learning models are more computationally intensive, they typically outperform machine learning models, which are lighter and faster to train and deploy. This study builds upon prior research studies demonstrating the effectiveness of deep learning models in DDoS attack detection.

Research studies such as P. Shorubiga et al. [2] and S. B. Mallampati et al. [5] have shown that CNN models can achieve high accuracy rates of 99% and 98.19%, respectively, indicating they are suitable for real-time intrusion detection. Similarly, A. I. Jony [3] attained 98.75% accuracy, demonstrating the capacity of recurrent models to capture temporal dependencies in network traffic. Additionally, J. Dalvi, et al. [1], achieved 89.62% accuracy in their experiment, suggesting that even simpler neural architectures can perform effectively while remaining computationally efficient. Building on these findings, this study develops a robust deep learning model that leverages the strengths of prior approaches to enable accurate and scalable DDoS detection. A Transformer-based model was implemented to classify network traffic as either benign or malicious. Unlike traditional neural networks, the Transformer employs a self-attention mechanism, allowing it to capture long-range dependencies within network traffic features. The model was trained using the Adam optimizer and binary cross-entropy loss, with dropout applied to mitigate overfitting. Performance evaluation was conducted using accuracy, precision, recall, and F1-score to provide a comprehensive assessment of detection effectiveness.

A Deep Neural Network (DNN)-based model was also implemented to compare its performance with the Transformer model. The dataset was preprocessed by encoding labels, normalizing feature values, and splitting the data into training and testing sets. The DNN architecture consisted of multiple dense layers with ReLU activation functions and dropout layers to mitigate overfitting, followed by a sigmoid output layer for binary classification. The network was trained using the Adam optimizer and binary cross-entropy loss function. Model performance was evaluated using accuracy, confusion matrix, and classification report metrics to ensure a comprehensive assessment. The objective of this research is to enhance DDoS detection accuracy beyond existing baselines through deep learning approaches. Three models—DNN, Transformer, and CNN—were implemented using the CIC-IoT2023 dataset to classify traffic as benign or malicious. In this phase, we introduced several methodological improvements, including merging DoS and DDoS into a single Flood class to reduce misclassification and applying targeted class balancing to address the dataset’s heavy imbalance. These adjustments helped stabilize training and improved the models’ ability to generalize. As a result, we successfully increased the

overall accuracy to 99% as well as the efficiency of all models, demonstrating measurable improvements over the initial results. Thus, establishing an effective methodology for detecting DDoS in IoT networks.

II. LITERATURE REVIEW

A. Group1: Deep learning models

As IoT devices have become increasingly widespread and vulnerable to attacks, researchers have turned to deep learning models such as CNNs, LSTMs, ANNs, DNNs, and RNNs to automatically detect intrusions in network traffic, often using benchmark datasets from the Canadian Institute for Cybersecurity (CIC) to conduct their experiments. These deep learning models have gained popularity because they can automatically learn patterns from network data without the need for manual feature selection, making them effective at identifying both known and new types of attacks, such as DDoS and intrusions. Thus, these approaches are capable of detecting both established and emerging cyberattacks, including distributed denial-of-service (DDoS), brute-force, and other intrusions.

These models often yield higher performance compared to traditional machine learning approaches and are particularly effective for building robust intrusion detection systems. In [2], P. Shorubiga and R. Shyam implemented a CNN to detect HTTP flooding DDoS attacks and achieved a 99% accuracy rate. Their findings demonstrate that CNN-based models can detect flooding attacks during their early stages, making them suitable for real-time intrusion detection. A similar trend is evident in [5], where S. B. Mallampati and S. Hari also applied a CNN and achieved an accuracy of 98.19%, further confirming the reliability of convolutional models for security tasks involving network traffic data. Collectively, these CNN-based studies indicate that convolutional models are particularly effective for capturing spatial patterns in network traffic and provide reliable performance across different datasets.

Other deep learning models have also shown promising performance. In [3], A. I. Jony and A. K. B. Arnob utilized an LSTM model and reported an accuracy of 98.75%, highlighting the benefit of using recurrent networks to capture temporal dependencies within network flows. Their work illustrates that models capable of learning time-based patterns can effectively identify intrusions that unfold over sequences of packets. Compared to CNNs, LSTMs offer the advantage of modeling sequential relationships in network data, which can be critical for detecting attacks that evolve. However, not all deep learning-based experiments achieve extremely high accuracy. For example, J. Dalvi et al. [1] employed a deep learning approach but obtained an accuracy of 89.62%. This outcome shows that performance can vary depending on the dataset, chosen architecture, and experimental setup, and that deep learning does not always guarantee near-perfect accuracy. Nonetheless, the collective findings across these works illustrate that deep learning continues to play a central role in advancing intrusion detection for IoT networks. Overall, the studies in this group demonstrate a clear trend: deep

learning models, particularly CNNs and LSTMs, consistently outperform traditional approaches in accuracy and adaptability.

B. Group2: Machine learning models

As IoT networks continue to expand, traditional machine learning (ML) techniques remain widely adopted for intrusion and attack detection due to their efficiency, interpretability, and suitability for deployment on resource-constrained devices. Unlike deep learning models, which automatically learn high-level representations from data, ML approaches typically rely on engineered features but offer faster training and inference times, making them practical for real-time or edge-level security applications. Across this group of studies, the models most frequently utilized include Logistic Regression (LR), Random Forest (RF), Decision Tree (DT), K-Nearest Neighbors (KNN), and Support Vector Machine (SVM). The CIC-IoT2023 dataset serves as the primary benchmark, providing IoT network traffic containing DDoS, DoS, reconnaissance, and other attack patterns. This dataset enables researchers to evaluate general-purpose ML classifiers on realistic IoT traffic and assess their effectiveness in detecting cyber threats.

In this study, K. R. Bommineni, et al. [10] evaluated Bagging-Decision Tree, AdaBoost, and KNN models for classifying DDoS traffic in multiclass scenarios. Among the tested models, Bagging-DT achieved an accuracy of 99.96%, demonstrating that lightweight ensemble techniques can deliver strong performance while maintaining low computational overhead. These results highlight that ensemble-based ML approaches offer a practical balance between detection accuracy and operational efficiency, especially for IoE systems requiring real-time analytics.

Similarly, K. Hasan et al. [11] explored the use of Random Forest, Decision Tree, and Logistic Regression within an SDN environment. Their experiments revealed that the Decision Tree model achieved 99.97% accuracy, surpassing the other classifiers and underscoring its suitability for rapid decision-making in dynamic network architectures like SDN. The study further illustrates that well-tuned ML models can provide real-time DDoS detection without the computational demands associated with deep learning.

Additional evidence of ML effectiveness is provided in [12], where SVM and Logistic Regression were deployed directly on edge devices. The SVM and LR models achieved true positive rates of 99.69% and 99.44%, respectively, correctly identifying nearly all DoS attack instances. By running ML algorithms at the edge, the system reduces communication overhead, enhances privacy, and improves response times—demonstrating the advantages of deploying lightweight ML directly on IoT hardware.

Finally, [13] compared several classifiers, including Decision Tree, Random Forest, Logistic Regression, AdaBoost, and SVM. Most models obtained accuracies above 80%, reinforcing the utility of ML techniques for securing IoT networks. However, the SVM classifier performed the worst, achieving 76.09% accuracy, illustrating that model performance can vary widely depending on the feature space, attack distribution, and

hyperparameter configuration. These findings highlight that while ML models generally perform well, no single model consistently dominates across all IoT scenarios.

Overall, the studies in this group show a clear pattern: traditional machine learning models, especially Decision Trees, Random Forests, and ensemble methods, provide strong performance for detecting IoT-based attacks such as DDoS, DoS, and reconnaissance. Their low computational cost and high accuracy make them practical for real-time intrusion detection on IoT and edge devices, where resource limits often prevent the use of more complex deep learning architectures.

C. Group3: Hybrid learning models - combination of multiple modeling approaches

As IoT networks continue to evolve and generate increasingly complex traffic patterns, hybrid learning approaches have gained prominence as a means of improving intrusion and DDoS attack detection. Unlike single-model techniques, hybrid methods integrate multiple machine learning, deep learning, or algorithmic frameworks to leverage the strengths of different modeling paradigms. As a result, the models explored in this group vary widely and include QRIME-SDPN, Support Vector Machine (SVM), XGBoost, Long Short-Term Memory (LSTM), Deep Neural Networks (DNN), K-Nearest Neighbors (KNN), Large Language Models (LLMs), AlexNet, Mini-VGGNet, and Random Forest. These combinations enable more flexible, accurate, and robust intrusion detection systems capable of handling diverse IoT attack patterns.

Most studies relied on the dataset CIC-IoT2023, and a smaller number of studies employed datasets such as CSE-CIC-IDS2018 and NSL-KDD. These datasets offer realistic and labeled network flows, enabling researchers to evaluate the performance of hybrid systems across normal and malicious IoT activity. Reported accuracy levels range from 78.24% to 99.99%, demonstrating that hybrid approaches consistently provide strong performance.

In [16], L. Liu, P. Wang, J. Lin, and L. Liu combined classical machine learning models such as Random Forest and XGBoost with LSTM networks to address intrusion detection in highly imbalanced traffic. Random Forest initially achieved the highest accuracy of 94.89%. However, the authors introduced the Difficult Set Sampling Technique (DSSTE), which significantly improved deep learning model performance and ultimately enabled the hybrid approach to outperform standalone ML models.

An advanced hybrid framework is seen in [19], where M. Saber, N. Gusarova, and V. Bogatyrev developed the Hybrid AutoML for Adaptive Learning Models (HAMI). This architecture integrates AutoML with adaptive online learning techniques by deploying hyperparameter-optimized models, such as Leveraging Bagging, Adaptive Random Forest, and Streaming Random Forest. HAMI achieved state-of-the-art performance, including an accuracy of 99.56%, demonstrating its capability to overcome the limitations of traditional intrusion detection systems by adapting to evolving IoT traffic patterns.

Another notable contribution is provided in [18] by S. Cheekati, C. R. Borra, S. K. V. R. V. Rayala, S. K. Sangula and V. Kulkarni, who proposed a hybrid QRIME-SDPN model combining quantum-inspired feature selection with deep learning based classification. Their system achieved an accuracy of 93.6% on the CIC-IoT2023 dataset, showing the promise of integrating advanced feature engineering strategies with deep neural classifiers to improve IoT threat detection.

Finally, in [20], A. Berqia, H. Bouijij, A. Merimi, and A. Ouaggane implemented a hybrid design using Logistic Regression, KNN, and DNN models to detect DDoS attacks within IoT environments. Their model achieved an accuracy of 99.99%, demonstrating that combining traditional ML and deep learning models can yield extremely precise results and an efficient system for identifying and classifying complex attack patterns. Overall, the studies in this group highlight a consistent trend: hybrid learning models provide enhanced adaptability, high accuracy, and strong generalization across diverse IoT attack scenarios. These results suggest that hybrid frameworks are a powerful direction for future IoT intrusion detection research.

III. METHODOLOGY AND BASELINE MODEL

This section describes the methodology used to implement and evaluate intrusion detection models on the CIC-IoT-2023 dataset. In this research, we reproduced the baseline models presented by Tseng et al. [26], including a Transformer-based architecture and a Deep Neural Network (DNN). In addition to replicating the baseline, we implemented a Convolutional Neural Network (CNN) to provide a broader comparison across deep learning architectures.

A. Data preparation

A. Data preprocessing

We utilized the CIC-IoT-2023 dataset developed by Canadian Institute for Cybersecurity. This dataset was selected because it is currently the largest publicly available IoT intrusion detection dataset and provides a comprehensive representation of IoT network traffic.

- 1) Data Merging and Sampling: The original dataset is distributed across 169 CSV files and contains 46,686,579 records. Because the dataset was updated after its initial release, we merged 63 available CSV files, yielding 45,018,243 samples, which represent 96.4% of the complete dataset.
- 2) Data Cleaning: A total of 1,000 rows containing invalid or missing values were removed, resulting in a final dataset of 45,018,243 clean samples (96.4% of the original data).
- 3) Class Merging: The original 34 attack categories were merged into 7 broader classes based on their underlying attack types. The table below provides an overview of the newly defined class groups.

Table 1: Class Distribution (CIC-IoT-2023 dataset)

- 1) Feature Preparation: The dataset contained 39 numerical features, including flow statistics, protocol fields, flag

Class	Samples Count
Flood	2,000,000
Mirai	2,019,703
Spoofing	373,731
Benign	841,050
BruteForce	10,018
Recon	528,887

counts, and other statistical measures. The baseline paper reports using 46 features; this discrepancy is likely attributable to differences in dataset versions or preprocessing choices across releases.

- 2) Data Normalization: All numerical features were standardized using the *StandardScaler*, which transforms each feature to have zero mean and unit variance.
- 3) Data Splitting: The dataset was divided into 80% for training (36,014,594 samples) and 20% for testing (9,003,649 samples). During training, 20% of the training partition was further allocated as a validation set.

B. Baseline Model

In the initial phase of this research, we replicated two baseline models proposed by Tseng et al. [26].

- 1) Transformer Model: The Transformer-based model was implemented to detect network intrusions and Distributed Denial of Service (DDoS) attacks using the CIC-IoT-2023 dataset. In accordance with the baseline study, three traffic categories were used for experimentation: Benign, HTTP Flood, and UDP Flood. The dataset was preprocessed by removing missing and infinite values, normalizing all numerical features using the *MinMaxScaler*, and encoding labels into numerical representations for classification. A Transformer Encoder architecture was employed due to its capability to capture temporal and contextual dependencies among network flow features through the self-attention mechanism. The model architecture consisted of the following components:
 - a) Input Embedding Layer: 39 input features projected to a 512-dimensional latent space.
 - b) Two Transformer Encoder Layers: Each containing one attention head and a 512-unit feedforward network.
 - c) Dropout Layers (0.1–0.2): Applied to mitigate overfitting.
 - d) Fully Connected Classifier: producing outputs corresponding to the three traffic categories.

The model was trained for 15 epochs using the Adam optimizer (learning rate = 0.001) with a batch size of 1,024. Focal Loss ($\alpha = 0.25$, $\gamma = 2.5$) was used to address the severe class imbalance present in the dataset, and class weights were computed and applied during training. Early stopping with a patience of five epochs was used to prevent overfitting, and the best model was saved based on validation accuracy. All training procedures were implemented in TensorFlow 2.15 using a system equipped with four NVIDIA RTX A5000 GPUs.

DNN Model: A Deep Neural Network (DNN) baseline model was implemented to provide a comparative benchmark against the Transformer architecture. The DNN was trained using the same preprocessed CIC-IoT-2023 dataset, which included all eight attack categories: Benign, DDoS, DoS, Mirai, Reconnaissance, Spoofing, Web-Based attacks, and BruteForce. The preprocessing pipeline remained consistent across models and included the removal of missing values, standardization of all numerical features using the *StandardScaler*, and label encoding for eight-class classification. A fully connected neural network architecture was selected due to its simplicity and computational efficiency. The model architecture consisted of the following components:

- 1) Input Layer: 39 network traffic features.
- 2) Three Hidden Layers: Each containing 256 neurons with ReLU activation.
- 3) Batch Normalization: Applied after each hidden layer to stabilize and accelerate training.
- 4) Dropout Layers (0.1): Incorporated to reduce overfitting.
- 5) Output Layer: Eight units with softmax activation for multi-class classification.
- 6) Total Parameters: Approximately 147,000 (approximately $11\times$ fewer than the Transformer model).

The model was trained for 15 epochs using the Adam optimizer (learning rate = 0.001) with a batch size of 1,024. Focal Loss was employed to address severe class imbalance, and class weights were computed to increase the importance of minority classes during training. Early stopping with a patience of five epochs was applied to prevent overfitting, and the best-performing model was saved according to validation accuracy. All experiments were conducted using TensorFlow 2.15 on a system equipped with four NVIDIA RTX A5000 GPUs.

C. Advanced Model

1. Advanced Model Improvements

To improve upon the baseline Transformer and DNN models and address classification challenges observed during initial experimentation, several enhancement strategies were developed. These improvements focused primarily on mitigating extreme class imbalance, reducing confusion between statistically similar attack types, and improving detection performance for minority classes that were consistently misclassified in the baseline results.

- 1) Baseline Transformer Tuning: The advancement phase began by refining the original Transformer model trained on the CIC-IoT-2023 dataset. The baseline implementation utilized global SMOTE and Focal Loss to address the heavy class imbalance across the eight attack categories. While this approach improved overall accuracy, significant confusion remained between classes with highly similar statistical profiles. To address these issues, the following strategies were implemented:
- 2) Manual Class Weight Adjustment: A targeted loss-weighting strategy was applied to resolve persistent misclassification between the Reconnaissance and DoS

categories. Both attacks share similar flow-level characteristics, particularly within layers 3 and 4 of the network stack, causing the baseline model to incorrectly predict DoS when presented with Recon samples. To correct this, manually tuned class weights were introduced into the loss function:

- 1) The loss contribution for the Reconnaissance class was increased to emphasize the minority class during training.
- 2) The weight assigned to DoS was reduced to prevent it from dominating the decision boundary.

This manual rebalancing forced the model to better separate the two classes and improved recall for Reconnaissance without sacrificing overall performance.

Hierarchical Attack Grouping: During error analysis, we observed that DDoS and DoS attacks are statistically indistinguishable in the CIC-IoT-2023 dataset. Their flow characteristics—including packet rate, duration, and byte count—overlap to such an extent that expecting a model to differentiate them is unrealistic from a data standpoint. Therefore, a hierarchical classification approach was introduced:

- 1) The DDoS and DoS labels were merged into a single super-class called Flood.
- 2) The classifier was retrained to distinguish Flood attacks from other categories rather than forcing a fine-grained distinction not supported by the dataset.

This modification reduced false positives between the two flood categories and stabilized the model’s decision boundaries, leading to more consistent performance across all classes.

Targeted SMOTE for Minority Classes: Even after applying global SMOTE, the BruteForce classes remained significantly underrepresented and were the most difficult for the model to detect. To address this issue, a targeted oversampling strategy was implemented:

- 1) SMOTE was applied exclusively to the BruteForce class, increasing its sample count from approximately 2,000 to 150,000.
- 2) All other classes retained their original sampling distributions.
- 3) The Flood class merge from Strategy B remained in place during this phase.

This targeted SMOTE application strengthened the model’s ability to learn minority-class decision boundaries without significantly altering the overall data distribution. As a result, recall and precision scores improved substantially for BruteForce and other low-frequency classes.

- 1) **CNN Model:** A Convolutional Neural Network (CNN) model was implemented to further expand the comparative analysis between traditional deep learning architectures and the baseline Transformer and DNN models. Similar to the previous models, the CNN was trained using the preprocessed CIC-IoT-2023 dataset, following the same data cleaning procedures, label encoding, and normalization steps to ensure consistency across all

experiments. Although CNNs are traditionally designed for image data, this model was adapted for tabular network traffic by reshaping feature vectors into a one-dimensional format suitable for Conv1D processing. The use of convolutional layers allows the model to learn local spatial feature patterns within sequences of network flow attributes, enabling effective representation learning even in non-image domains such as IoT traffic analysis. The architecture consisted of three sequential convolutional blocks with increasing filter sizes to progressively capture higher-level representations from the input features. Each block contained the following components:

- a) **Conv1D Layers:** With 128, 256, and 512 filters, respectively, each followed by a ReLU activation to introduce non-linearity.
- b) **Batch Normalization:** Applied after each convolutional layer to stabilize learning and reduce internal covariate shift.
- c) **Dropout Layers:** Incorporated for regularization and to prevent overfitting.
- d) **MaxPooling Layers:** These were used after each convolutional block to downsample the feature maps and reduce computational complexity.

Following the convolutional blocks, a Global Average Pooling layer was applied to convert the final feature maps into a vector representation, significantly reducing dimensionality while retaining spatial feature information. A fully connected dense layer was added before the output to learn high-level interactions between the extracted features. The final output layer produced class probabilities corresponding to the target attack categories. The training configuration for the CNN model was aligned with the baseline models to ensure a fair comparison. The model was trained using the same training function and identical hyperparameters, including a batch size of 1,024, a learning rate of 0.001, and a training schedule of 15 epochs. Focal Loss was again employed to mitigate the impact of severe class imbalance, and class weights were computed and applied during training. Early stopping with a patience of five epochs was used to prevent overfitting, and the best-performing model was saved based on validation accuracy. All training procedures were implemented using TensorFlow 2.15 on a system equipped with four NVIDIA RTX A5000 GPUs.

D. Evaluation Metrics

All models were evaluated on the test set using a set of performance metrics. The primary evaluation metrics consist of the following:

- 1) **Accuracy:** Overall classification accuracy across all classes
- 2) **Per-class metrics:** Precision, recall, and F1-score computed individually for each of the eight attack categories
- 3) **Confusion matrix:** It is used to identify frequently misclassified attack pairs and examine class-specific error patterns

- 4) Macro and weighted averages: It is employed to assess balanced performance across classes, particularly in the presence of severe class imbalance

IV. RESULTS

A. Overall Performance

The initial results from the baseline models demonstrated comparable performance on the CIC-IoT-2023 dataset. The Transformer model achieved a testing accuracy of 85.26%, while the DNN baseline achieved 85.46%, reflecting a marginal difference of 0.20%. Precision, recall, and F1-scores across the eight classes indicated strong generalization, with both models correctly identifying the majority of attack categories while maintaining reliable performance on benign traffic. Following hyperparameter tuning and optimization, the performance of both baseline models and the newly implemented CNN model improved significantly, achieving approximately 99% accuracy across all models. Specifically, the Transformer achieved 98.97%, the DNN achieved 99.09%, and the CNN achieved 99.08% accuracy. These results are comparable to those reported in [26], where the Transformer achieved 99.40% and the DNN achieved 99.36% accuracy. Overall, these results indicate that the models have become more advanced, demonstrating substantial improvement from the initial phase and the ability to detect DDoS attacks in both HTTP and HTTPS traffic. A comparative summary of model performance during the initial and final phases is presented in the tables below.

Model	Accuracy (%)	Parameters	Macro F1	Training Time/Epoch
Transformer	98.97	1,602,568	0.7232	~ 50 seconds
DNN	99.09	146,952	0.7452	~ 30 seconds
CNN	99.08	630,535	0.7447	~ 62 seconds

Table 2: Comparative Analysis of the Baseline Models

B. Per-Class Performance

Our improved Transformer and DNN models achieved greater accuracy in Mirai, Reconnaissance, Web-Based, Spoofing, and Brute Force than [26]. This may be due to the use of targeted SMOTE for the underrepresented classes. Combining DoS and DDoS into one class led to a large spike in accuracy from 28.6% for DDoS in [26] to 100%. The merge and downsampling made it easier for our models to differentiate DoS and DDoS not only from each other, but from other attack labels as well. Overall, our Transformer and DNN models improved accuracy in every class when compared to [26]. The tables below detail the results of each classification.

Class	Transformer (%)	DNN (%)	Number of Samples
Benign	80.00	81.36	2,306
BruteForce	75.00	88.89	28
Flood (DDoS + DoS)	100.00	100.00	89,391
Mirai	98.75	98.96	5,630
Reconnaissance	79.54	80.10	1,528
Spoofing	89.99	92.45	1,072
Web-Based	55.56	85.71	45

Table 3: Per-Class Performance of Advanced Models

C. Comparison with Published Results

Although accuracy increased for each individual class, the overall accuracy fell just short of [26]. Tseng et al. [26] reported an accuracy of 99.40% for the Transformer model and 99.36% for DNN. Our reported accuracy is 98.97% for the Transformer and 99.09% for DNN. This may be due to the limited time we had to work on the models. The tables below present a comparison of our implementation and the baseline study in [26] for both the initial and final phases.

Model	Dataset	Features	Accuracy (%)	Gap (%)
Transformer	45M (96%)	39	98.97	0.43
Tseng et al. [26]	46.7M	46	99.40	–
DNN	45M (96%)	39	99.09	0.27
Tseng et al. [26]	46.7M	46	99.36	–
CNN	45M (96%)	39	99.08	0.28

Table 4: Comparative Analysis with Published Results

In summary, our models achieved approximately 85% accuracy during the initial phase, and after tuning, their performance increased to approximately 99%, compared to Tseng et al. [26], who reported accuracies of 99.40% and 99.36%. This reflects a performance gap of roughly 0.2–0.4%. Several factors may contribute to this difference:

- 1) Class Merging: We merged the DDoS and DoS attacks into a single category (“Flood”). This consolidation reduces class granularity and may improve overall performance by combining attacks with similar behavioral patterns.
- 2) Feature Discrepancy: Our implementation used 39 features, whereas the baseline study used 46 features (a 15% reduction). The cause of this discrepancy is unclear and may influence classification accuracy.
- 3) Training Strategy: We employed Focal Loss and class weights to address severe class imbalance, whereas the baseline study used standard cross-entropy. This difference may affect the accuracy–recall tradeoff.
- 4) Implementation Details: Variations in hyperparameters, weight initialization, and preprocessing steps—some of which were not fully documented in the baseline paper—may also contribute to the observed performance gap.

Despite this small gap, our models demonstrate strong intrusion detection capability, achieving 99% accuracy and near-perfect performance on web-based attacks. The consistent results across the Transformer, DNN, and CNN models indicate that both architectures effectively learned the underlying attack patterns. Future work focusing on the feature discrepancy and further hyperparameter optimization may help close the remaining performance gap.

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